

EE 231: Measurement and Instrumentation

Course Code	EE 231
Course Name	Measurement and Instrumentation
L-T-P-C	2-0-2-3
Prerequisites	IC161 – Applied Electronics, EE261 - Electrical Systems around us, EE260 – Signals and Systems.
Intended for	B.Tech. Electrical Engineering, allied B.Tech programmes
Approval	56th BoA (Jul 2024)

Course Contents

- **Fundamentals of measurement:** Physical quantities, dimensional analysis, significant figure calculations, errors in measurement, Taguchi method. (2 Lectures)
- **Measurement foundations:** Measurement of resistances – wire and bridge methods; Measurement of impedances – bridge methods; analog and digital meters; the digital storage oscilloscope. (4 Lectures)
- **Sensors and Actuators:** Sensors – Definitions, classification, static and dynamic characteristics, sensor examples, e.g. temperature, pressure, piezoresistive, piezoelectric, capacitive, etc.; Actuators – e.g. piezoelectric actuator, DC motors, servo motor, stepper motor. (8 Lectures)
- **Instrumentation:** Signal conditioning – amplification, filtering, isolation and handling noise; Power supply and regulation essentials; Data acquisition approaches – principles of ADC-based acquisition, use of DAQ cards; Calibration principles and reference to standards; Standard computer interfaces and communication protocols. (8 Lectures)
- **System-level design:** Elements of system-level thinking; introduction to microcontrollers and FPGAs; Integration of sensors, signal conditioning and data acquisition systems; Problem-based learning based on real-world examples – e.g. distance sensors, PID controllers, real-time data acquisition systems, etc.; Troubleshooting strategies. (6 Lectures)

Laboratory Modules

1. **Instrumentation fundamentals:** soldering; using power sources, signal generators, oscilloscopes; SCPI programming (2 hours).
2. **The Wheatstone Bridge:** First principles calibration (1 hour).
3. **The Instrumentation amplifier:** Verification of working, amplification of Wheatstone bridge output (1 hour).
4. **The ADC:** use of off-the-shelf ADCs (ADS1115s, or MCP3008) with standard microcontrollers (e.g. Arduino) for data acquisition (1 hour).
5. **System design:** an end-to-end temperature logger using a thermistor. (2 hours)
6. Real-time temperature stabilization of a Peltier element using PID control and the temperature logger (2 hours).
7. Elements of digital signal processing (2 hours).
8. PID control of speed of a DC motor (3 hours).

Text Books

1. A. K. Sawhney, **A course in Electrical and Electronic Measurements and Instrumentation**, Dhanpat Rai and Co.

References

1. Doebelin, E. O., Manik, D. N., **Measurement Systems**, 6th Edition, Tata McGraw Hill India, 2011.
2. Fraden, Jacob., **Handbook of modern sensors**, Springer Science+Business Media, 2010.